IBM DATATHON 2026

System Architecture

INPUT LAYER

Commercial satellite operators + authorized sovereign or defense data sources

Authenticated, encrypted telemetry feeds

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CONFIDENTIAL COMPUTING LAYER

IBM LinuxONE / IBM z16 with Secure Execution capabilities

Isolated processing of protected workloads

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PROCESSING CORE

Orbital propagation, closest-approach estimation and collision-risk analysis

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OUTPUT LAYER

Risk alerts, proposed maneuver vectors and secure operator notifications

ARCHITECTURE OVERVIEW

- **End-to-End Security:** Ingestion of authenticated telemetry feeds directly into isolated, hardware-enforced confidential computing environments.
- **High-Performance Analytics:** Real-time orbital estimation and collision-risk modeling running on IBM LinuxONE / z16 infrastructure.
- **Actionable Insights:** Automated output generation delivering high-priority risk alerts and maneuver recommendations.